Alexis Valencia

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EDUCATION

Carnegie Mellon University (CMU) **Pittsburgh, PA**
Master of Science in Computational Biomedical Engineering – Research | GPA: 3.67/4.0 May 2026
Relevant Coursework: Brain Computer Interface, Neural Signal Processing, Fundamentals to Computational BME
University of California – Riverside (UCR) **Riverside, CA**
Bachelor of Science in Bioengineering | GPA: 3.58/4.0 June 2024
Relevant Coursework: Biosystems and Signal Analysis, Systems Neuroscience

SKILLS

Programming: C/C++, MATLAB, Python, C#, JavaScript
Software: GitHub, Fusion 360, Altium Designer, Cura, Adobe Illustrator, Unity (XR/AR/VR), Microsoft Office
Embedded Systems: STM32, Arduino, SPI, DMA, UART, ADC Sampling, Real-Time Data Acquisition, Peripheral Interfacing, PCB Design, Signal Acquisition
Biomedical Systems: EEG, CW-NIRS, Signal Processing, Time-Series Analysis, Feature Extraction, Classification
Hardware: 3D printing (FDM/SLA), Soldering, Rapid Prototyping, Oscilloscope, Logic Analyzer, Multimeter
Design & Fabrication: CAD Modeling, Rapid Prototyping, FDM/SLA Printing, Laser Cutting, Physical Prototyping, Product Iteration, Design Validation

RESEARCH EXPERIENCE

Graduate Research Assistant July 2024 – May 2026
Wearable Near-Infrared Spectroscopy **|** Biophotonics Lab at CMU

- Designed and prototyped a wearable embedded sensing system integrating flexible PCB design, optical sensing, and real-time data acquisition for continuous brain monitoring
- Arranged and validated PCB layouts and embedded hardware prototypes using Altium Designer for wearable sensing systems
- Developed a **wearable optical embedded system** to enable continuous non-invasive brain monitoring
- Developed embedded firmware in C for STM32-based biosignal acquisition systems using SPI and DMR-driven sampling workflows
- Iteratively refined hardware prototypes through rapid testing, debugging, and performance validation
- Debugged and validated hardware communication and timing behavior for real-time optical signal acquisition
- Performed low-level hardware integration including sensor interfacing, ADC sampling, and peripheral communication
- Improved signal-to-noise ratio (SNR) by **~2 dB** through hardware and signal optimization, enabling more reliable physiological measurements
- Engineered **two custom MATLAB data pipelines** (OOP-based) for automated parsing, filtering, and analysis of optical signals
- Designed and validated a **flexible PCB (Altium Designer)**, optimizing photodetector selection and AFE performance
- Implemented real-time data acquisition pipelines using SPI and DMA to support high-throughput biosignal sampling on STM32 microcontrollers
- Integrated **microcontroller-based data acquisition (SPI/DMA workflows)** for real-time bio-signal sampling

Undergraduate Research Assistant June 2023 – June 2024
Augmented Reality Experiment for Schizophrenia Study **|** Systems Neural Engineering Laboratory at UCR

- Led the development of an **augmented reality experimental platform** for behavioral studies in schizophrenia
- Built an interactive environment in **Unity (C#)** deployed on **Magic Leap 2**, including hand tracking and task-based interaction

PROFESSIONAL EXPERIENCE

AR/VR Developer | XCITE Center for Teaching and Learning, UCR March 2023 – June 2024

- Designed and developed interactive educational XR experiences using Unity for Meta Quest and mobile platforms to teach abstract engineering and encryption concepts
- Developed a **cross-platform XR educational application** (VR+AR) using Unity for Meta Quest and mobile devices
- Led rapid prototyping and development of an interactive educational game environment translating cryptography concepts into hands-on visual learning experiences
- Led a team of **three** in building an **interactive encryption learning environment**, enhancing student engagement through immersive simulations
- Designed and implemented **interactive 3D systems (e.g., Enigma machine)** using Fusion 360 + Unity (C# scripting)
- Delivered live demos to **60+ students**, collecting usability feedback and iterating on user experience

- Iterated on gameplay mechanics and user interaction based on live classroom demonstrations and student usability feedback
- Created physical-to-digital educational assets through CAD modeling, 3D design, and interactive simulation workflows
- Collaborated with **four** professors in revamping student education by translating curriculum into interactive digital learning tools
- Hosted a high school summer camp by teaching **thirteen** students how to print 3D models and design virtual assets

3D Printer Specialist | Clever Minds Tutoring Center, Laredo, TX June 2020 – August 2023

- Operated, maintained, and optimized multiple additive manufacturing systems for rapid prototyping and educational fabrication workflows
- Maintained and optimized **four** FDM 3D printing systems, reducing downtime and improving print reliability
- Troubleshot print failures, material issues, and hardware inconsistencies to improve fabrication reliability
- Assisted in transforming digital CAD concepts into physical functional prototypes
- Trained **three** staff on end-to-end 3D printing workflow, from online CAD model to printed toys for children
- Supported STEM education initiatives through hands-on engineering instruction

SELECTED PROJECTS

Improving 2-D Motor Imagery BCI Classification | Course Project, CMU October 2025 – December 2025

- Conducted EEG-based motor imagery experiments and analyzed electrophysiology for machine learning classification
- Built a signal processing pipeline using **bandpass filtering** + **Common Spatial Patterns (CSP)**
- Trained a **Random Forest classifier**, improving F1-score from 50.9% to 78.8%
- Implemented machine learning and signal analysis workflows in Python using scientific computing libraries

Optical Coherence Tomography and Vibrometry Endoscope | Capstone Project, UCR September 2023 – June 2024

- Co-designed and manufactured a **compact OCT-based diagnostic device** for assessing hearing loss
- Engineered **seven** optomechanical components with micro-adjustment mechanisms, improving stability and alignment
- Reduced system weight while maintaining consistent optical return power over extended use